

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1361

Milagro 138 kV
January 15, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1361. The Interconnection Customer proposes a 68.4 MW battery energy storage facility interconnecting at the Milagro 138 kV substation in the Virginia area. The request was received on December 19, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,091,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Carmel 146Th St to Oak Creek 138.0 kV L
Contingency: Bear to Clover 345.0 kV line
Loading: 166.2% (Rating: 200 MVA)
DFAX: 5.0%
Pre-project loading: 164.5%
2. Facility: Ceiba Baja 7012 to Carmel 146Th St 138.0 kV Line
Contingency: Bear to Clover 345.0 kV
Loading: 164.8% (Rating: 204 MVA)
DFAX: 5.0%
Pre-project loading: 163.1%
3. Facility: Tangy Substation to Ceiba Baja 7012 138.0 kV
Contingency: Bear to Clover 345.0 kV line
Loading: 152.1% (Rating: 229 MVA)
DFAX: 5.0%
Pre-project loading: 150.6%
4. Facility: Michigan Power Lp to Osceola 138.0 kV
Contingency: Aspen to Lowmoor 345.0 kV
Loading: 111.0% (Rating: 355 MVA)
DFAX: 6.2%
Pre-project loading: 109.8%
5. Facility: Lone Tree to Osceola 138.0 kV L
Contingency: Aspen to Lowmoor 345.0 kV
Loading: 110.9% (Rating: 361 MVA)
DFAX: 6.2%
Pre-project loading: 109.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carmel 146Th St to Oak Crk 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,943,000	73% (\$4,330,000)
Ceiba Baja 7012 to Carmel 146Th St 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,536,000	73% (\$4,761,000)
Tangy to Ceiba Baja 7012 138.0 kV Line	Rebuild line with higher-capacity conductor	\$8,001,000	73% (\$5,829,000)
Michigan Power Lp Substation to Osceola 138.0 kV line	Rebuild line with higher-capacity conductor	\$11,822,000	1% (\$124,000)
Osceola to Lone Tree 138.0 kV	Rebuild line with higher-capacity conductor	\$4,464,000	1% (\$47,000)
	Total allocated to project		\$15,091,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.